

To: The Editor-in-Chief
Measurement and Control

From:

Zhuang Shao^{1,*}, Lijun Lei², Peng Wang³, Liang Zheng², Jie Zhou²

¹China Resources Power Technology Research Institute Co., Ltd., Shenzhen 518000, Guangdong Province, China

²Rundian Energy Science and Technology Co., Ltd., Zhengzhou 450052, Henan Province, China

³North China University of Water Resources and Electric Power, Zhengzhou 450046, Henan Province, China

*Corresponding author: shaozhuang@crpower.com.cn, ORCID: 0000-0003-2496-0797

Re: Submission of manuscript

“A Tunable GP-HOCBF Safety Filter for Supercritical Boiler-Turbine Control under Model Mismatch”

Dear Editor,

We submit the enclosed manuscript for consideration at Measurement and Control. The paper presents an industrial measurement-and-control safety filtering framework that preserves pressure, enthalpy, and power output constraints during supercritical power plant operation under model mismatch.

Fit with Measurement and Control. The journal’s scope encompasses measurement, control, and automation in industrial applications. Our work directly addresses this scope along four dimensions:

1. **Industrial measurement and control relevance.** Supercritical boiler-turbine units require continuous monitoring and constraint-preserving control of steam pressure, separator enthalpy, and power output. Model mismatch from coal quality variation, equipment aging, and unmodeled operating disturbances is a common commissioning concern. Our safety filter provides a real-time mechanism for maintaining these industrial operating constraints.
2. **Real-time safety filtering.** The method functions as an online safety projection layer inserted downstream of any upstream controller (LQR, MPC, PID, or a learned policy) without modifying the existing control architecture. Per-step safety-filter pipeline latency is ~ 25 ms (GPU, JIT-compiled), including controller action generation, GP inference, and QP solve, and is compatible with the 1 s sampling rate used in the benchmark loop.
3. **Constraint-preserving operation under model mismatch.** Under coal quality drift, fouling, valve degradation, and state-dependent coupling, nominal safety filters may certify unsafe actions as safe. GP mean correction reduces HOCBF violations by more than 500 times and achieves $\leq 0.18\%$ violation in six of seven uncertainty scenarios.
4. **Commissioning-calibrated robustness.** The method quantifies model uncertainty through Gaussian process residual learning with compositional uncertainty propagation, but treats the robustness factor ϵ_K as a tunable commissioning parameter rather than a fixed worst-case constant.

Key contributions.

1. An industrial safety-filter architecture for a 1000 MW supercritical boiler-turbine measurement and control loop.
2. Compositional GP uncertainty propagation through the HOCBF ψ -chain, yielding a per-constraint state-dependent margin with a high-probability forward-invariance certificate under fixed calibrated-GP assumptions.
3. A practical deployment envelope for the robustness factor ϵ_κ : zero margin for additive mismatch, a small positive margin for moderate state-dependent coupling, and fallback diagnostics beyond the envelope.
4. Validation on a 5th-order supercritical boiler-turbine benchmark with Φ -scaled nonlinear dynamics, eight controller/filter variants, six disturbance scenarios, and per-step latency of ~ 25 ms, approximately ten times faster than the implemented SLSQP-based NMPC baseline.

Validation scope and transparency. The manuscript reports simulation-benchmark validation and does not claim plant deployment. We state this boundary explicitly, provide reproducibility materials through the project repository, and identify hardware-in-the-loop validation and plant historian calibration as required next steps before field use.

Conflicts of interest. The authors declare no competing interests. The manuscript has not been published elsewhere and is not under simultaneous consideration by any other journal.

Author approval and writing assistance. All authors have read and approved the manuscript and consent to submission to Measurement and Control. During preparation of the manuscript, the authors used ChatGPT for language polishing and manuscript formatting. The authors reviewed and edited the content and take full responsibility for the submitted work.

Sincerely,

Zhuang Shao (on behalf of all authors)
shaozhuang@crpower.com.cn